1 Problem setup, identification and inference

1.1 Setup

A *response set* is a finite collection $U = \{1, \dots, N\}$ of records. Record i carries an outcome y_i, covariates x_i that include a detector score with flag $d_i \in \{0, 1\}$, and an origin a_i, the share of the response that is machine-written ($a_i \in \{0, 1\}$ unless stated otherwise; $a_i \in [0, 1]$ for hybrid authorship). Write $h_i = 1 - a_i$ for the human share, $N_H = \sum_U h_i > 0$, $N_A = N - N_H$ and $\varepsilon = N_A/N$. The estimand is the human mean of the response set,

$$\mu_H = \frac{\sum_{i \in U} h_i \, y_i}{\sum_{i \in U} h_i}, \tag{1}$$

and μ_A is the corresponding machine mean when $N_A > 0$. Outcomes, covariates and flags are observed for every record; origins are not. An *audit* verifies the origin of a subset $S \subset U$ of m records. Strata are indexed by g, with N_g records of which m_g are audited.

Assumption 1 (D1, probability audit)**.** S is drawn by a randomised design chosen by the analyst, with inclusion probabilities $\rho_i = \Pr(i \in S) > 0$ for every $i \in U$, known for the audited records. (Designs with several stages are treated in Lemma 2 with stage-wise probabilities.)

Assumption 2 (D2, correct verification)**.** For $i \in S$ the audit returns a_i. (Proposition 6 relaxes this.)

Assumption 3 (D3, complete first phase)**.** (y_i, x_i, d_i) is observed for every $i \in U$.

Assumption 4 (M1, non-differential flag)**.** Within each origin class the flag is uncorrelated with the outcome: $\mathrm{Cov}(D, Y \mid A = a) = 0$ for $a \in \{0, 1\}$.

D1–D3 are properties of the audit, which the analyst controls. M1 is a property of the detector and the outcome; without verified origins it can sometimes be refuted but never verified. It was the identifying condition of the first version of this paper. Below it is needed only for the moment inversion (Proposition 1); the estimators we recommend do not use it.

In Section 1.2 capital letters (Y, A, D) denote a draw from any joint distribution of outcome, binary origin and flag, in particular the empirical distribution of a response set, in which case $\Pr(A{=}1) = \varepsilon$, $\mathbb{E}[Y \mid A{=}0] = \mu_H$ and the identities hold exactly on that response set. Write $\bar{f} = \Pr(D{=}1 \mid A{=}0)$ for the false-positive rate, $\bar{s} = \Pr(D{=}1 \mid A{=}1)$ for the sensitivity and $C_a = \mathrm{Cov}(D, Y \mid A = a)$. A product of zero and an undefined conditional moment is read as zero.

1.2 Why deletion and moment inversion are biased

The naive mean has bias $\varepsilon(\mu_A - \mu_H)$. Deleting flagged records gives $\mathbb{E}[Y \mid D{=}0]$.

Lemma 1 (Deletion bias)**.** *Let* $\bar{f} < 1$ *and* $\varepsilon_{\mathrm{res}} = \Pr(A{=}1 \mid D{=}0) = \varepsilon(1 - \bar{s})/\{(1 - \varepsilon)(1 - \bar{f}) + \varepsilon(1 - \bar{s})\}$*. Then*

$$\mathbb{E}[Y \mid D{=}0] - \mu_H = \varepsilon_{\mathrm{res}} \, \Delta_{\mathrm{surv}} - \frac{C_0}{1 - \bar{f}}, \qquad \Delta_{\mathrm{surv}} = \mathbb{E}[Y \mid A{=}1, D{=}0] - \mathbb{E}[Y \mid A{=}0, D{=}0]. \tag{2}$$

Proof. By total probability over the origin of the unflagged records, $\mathbb{E}[Y \mid D{=}0] = \mathbb{E}[Y \mid A{=}0, D{=}0] + \varepsilon_{\mathrm{res}}\Delta_{\mathrm{surv}}$. Among humans, $\mathbb{E}[Y(1 - D) \mid A{=}0] = \mu_H(1 - \bar{f}) - C_0$, so $\mathbb{E}[Y \mid A{=}0, D{=}0] = \mu_H - C_0/(1 - \bar{f})$. □

The first term is residual contamination: machine records that survive deletion, times their outcome gap to the surviving humans. The second is a selection effect on the humans themselves: if falsely flagged people answer differently from other people ($C_0 \neq 0$), deleting them moves

the mean even when the detector catches every machine record. With a style stratum G and $f(G) = \Pr(D{=}1 \mid A{=}0, G)$, $C_0 = \mathrm{Cov}(f(G), \mathbb{E}[Y \mid A{=}0, G]) + \mathbb{E}[\mathrm{Cov}(D, Y \mid A{=}0, G)]$; the first submission assumed the last term away.

Proposition 1 (Moment inversion under M1)**.** *Let $0 < \varepsilon < 1$ and $\bar{s} \neq \bar{f}$, with $(\bar{s}, \bar{f}, \varepsilon)$ known. The system*

$$\mathbb{E}[YD] = (1-\varepsilon)\bar{f}\,\theta_H + \varepsilon\bar{s}\,\theta_A, \qquad \mathbb{E}[Y(1-D)] = (1-\varepsilon)(1-\bar{f})\,\theta_H + \varepsilon(1-\bar{s})\,\theta_A \tag{3}$$

in the unknowns (θ_H, θ_A) has a unique solution $(\tilde{\mu}_H, \tilde{\mu}_A)$, and under M1 $\tilde{\mu}_H = \mu_H$.

Proposition 2 (Bias of the moment inversion without M1)**.** *Under the conditions of Proposition 1 but without M1,*

$$\tilde{\mu}_H = \mu_H - \frac{C_0 + \frac{\varepsilon}{1-\varepsilon} C_1}{\bar{s} - \bar{f}}. \tag{4}$$

Proof of Propositions 1 and 2. The determinant of (3) is $\varepsilon(1-\varepsilon)(\bar{f} - \bar{s}) \neq 0$ and Cramer's rule gives $\tilde{\mu}_H = \{(1-\bar{s})\mathbb{E}[YD] - \bar{s}\,\mathbb{E}[Y(1-D)]\}/\{(1-\varepsilon)(\bar{f} - \bar{s})\}$. In truth $\mathbb{E}[YD] = (1-\varepsilon)(\bar{f}\mu_H + C_0) + \varepsilon(\bar{s}\mu_A + C_1)$ and $\mathbb{E}[Y(1-D)] = (1-\varepsilon)\{(1-\bar{f})\mu_H - C_0\} + \varepsilon\{(1-\bar{s})\mu_A - C_1\}$. Substituting, the terms in μ_A cancel and the numerator equals $(1-\varepsilon)\{(\bar{f} - \bar{s})\mu_H + C_0\} + \varepsilon C_1$, which gives (4); under M1, $C_0 = C_1 = 0$. □

The inversion is therefore biased whenever $C_0 + \frac{\varepsilon}{1-\varepsilon} C_1 \neq 0$, and for a given violation the bias is inversely proportional to the detector's separation $\bar{s} - \bar{f}$, the analogue of a weak-instrument bias.

1.3 Design-based identification

For a function z on U write $t_z = \sum_U z_i$ and $\hat{t}_z = \sum_S z_i/\rho_i$ for its Horvitz–Thompson estimator; under D1, $\mathbb{E}_p[\hat{t}_z] = t_z$ for every z that does not depend on S, where $\mathbb{E}_p$ is expectation over the audit design with the response set held fixed.

Definition 1 (Difference-ratio family)**.** Let $\eta : U \to \mathbb{R}$ be a predictor of the origin, computed from (x_i, d_i), and $\lambda \in \mathbb{R}$. Put

$$\hat{T}_\lambda = \lambda \sum_{i \in U} y_i(1-\eta_i) + \sum_{i \in S} \frac{y_i\{h_i - \lambda(1-\eta_i)\}}{\rho_i}, \qquad \hat{H}_\lambda = \lambda \sum_{i \in U}(1-\eta_i) + \sum_{i \in S} \frac{h_i - \lambda(1-\eta_i)}{\rho_i}, \tag{5}$$

and $\hat{\mu} = \hat{T}_\lambda/\hat{H}_\lambda$, reported when $\hat{H}_\lambda > 0$ (for a tuned member, also at $\lambda = 1$) and the audit contains at least two human records.

The family contains the two-phase Hájek estimator ($\lambda = 0$), flag-PPI ($\eta_i = d_i$, $\lambda = 1$), RC-PPI (η_i a fitted probability that record i is machine-written, $\lambda = 1$) and their power-tuned versions (flag-PPI++, RC-PPI++; a data-chosen λ).

Theorem 1 (Identification without M1)**.** *Assume D1–D3, with origins in $[0, 1]$, and let η and λ not depend on S. Then*

1. *$\mathbb{E}_p[\hat{T}_\lambda] = N_H\,\mu_H$ and $\mathbb{E}_p[\hat{H}_\lambda] = N_H$;*
2. *whenever $\hat{H}_\lambda \neq 0$,*

$$\hat{\mu} - \mu_H = \frac{\hat{t}_e - t_e}{\hat{H}_\lambda}, \qquad e_i = \{h_i - \lambda(1-\eta_i)\}(y_i - \mu_H). \tag{6}$$

No assumption links the predictor, the flag or the outcome: M1 is not used and η may be arbitrarily wrong.

Proof. (1) The sums over S are Horvitz–Thompson estimators of $\sum_U y_i\{h_i - \lambda(1-\eta_i)\}$ and $\sum_U\{h_i - \lambda(1-\eta_i)\}$; adding the first-phase sums leaves $\sum_U y_i h_i = N_H\mu_H$ and $\sum_U h_i = N_H$. (2) $\hat{T}_\lambda - \mu_H\hat{H}_\lambda = \lambda\sum_U(y_i-\mu_H)(1-\eta_i) + \hat{t}_e$, and $t_e = \sum_U h_i(y_i-\mu_H) - \lambda\sum_U(1-\eta_i)(y_i-\mu_H) = -\lambda\sum_U(1-\eta_i)(y_i-\mu_H)$ because $\sum_U h_i(y_i-\mu_H) = 0$ by (1). □

The human mean is thus identified by the audit design alone: numerator and denominator are design-unbiased whatever the detector does, and the error of the ratio is the Horvitz–Thompson error of the residual e. The detector enters only through e. For $\lambda = 1$ the residual is $(\eta_i - a_i)(y_i - \mu_H)$, the prediction error times the outcome deviation, which vanishes for a perfect predictor. Two qualifications apply. First, $\hat{\mu}$ is a ratio and has the usual ratio bias of order $1/m$, which is not zero in small audits. Second, the theorem is unconditional: an estimator that is reported only when the audit contains enough records of each class (as RC-PPI must be, since its predictor is fitted on the audit) is conditioned on an event that depends on the origins, and the theorem does not cover that conditional distribution. For this reason the primary (intention-to-treat) analyses score a complete rule (RC-PPI++ when feasible, flag-PPI on the same audit otherwise) over all replications; analyses restricted to feasible replications are conditional and are not covered by Theorem 1.

1.4 Variance, intervals and efficiency

Let $u_i = h_i(y_i - \mu_H)$ and $v_i(\eta) = (1-\eta_i)(y_i - \mu_H)$, so that $e = u - \lambda v(\eta)$ and $t_u = 0$.

Theorem 2 (Linearised variance and normal limit)**.** *Assume D1–D3 and consider a sequence of response sets with single-stage stratified simple random sampling without replacement, $m \to \infty$ and $m_g \geq 2$, along which (i) N_H/N is bounded away from zero, y and all predictions are bounded, and no stratum dominates the variance: $m\,N^{-2}\sum_g N_g^2/m_g$ is bounded; (ii) there are fixed η^* and λ^* such that, with $e^* = u - \lambda^* v(\eta^*)$, the Horvitz–Thompson total $\hat{t}_{e^*}$ satisfies a central limit theorem under the design, the stratified variance estimator $\hat{V}(e^*)$ is ratio-consistent for its variance, and the limit is not degenerate: $\liminf m\,N^{-2}\operatorname{Var}_p(\hat{t}_{e^*}) > 0$; (iii) the fitted predictor and tuning constant are stable: each fitted model $\hat{\eta}$ satisfies $N^{-1}\sum_U(\hat{\eta}_i - \eta_i^*)^2 \to 0$, the predictions $\tilde{\eta}$ used for the audited records satisfy $N^{-1}\sum_S(\tilde{\eta}_i - \eta_i^*)^2/\rho_i \to 0$, and $\hat{\lambda} \to \lambda^*$, all in probability; (iv) every prediction used for an audited record, in both sums of (5), comes from a model fitted without that record (cross-fitting), and $N^{-1}m^{1/2}\{\hat{t}_{v(\bar{\eta})-v(\eta^*)} - t_{v(\bar{\eta})-v(\eta^*)}\} \to 0$ in probability, where $\bar{\eta}$ denotes the predictions used in the first-phase sum ($\bar{\eta}_i = \tilde{\eta}_i$ for $i \in S$). Then $(\hat{\mu} - \mu_H)/\widehat{\mathrm{se}} \to \mathcal{N}(0,1)$ in distribution, with*

$$\widehat{\mathrm{se}}^{\,2} = \frac{\hat{V}(\hat{e})}{\hat{H}_\lambda^2}, \qquad \hat{e}_i = \{h_i - \hat{\lambda}(1-\tilde{\eta}_i)\}(y_i - \hat{\mu}), \qquad \hat{V}(\hat{e}) = \sum_g N_g^2\Big(1 - \frac{m_g}{N_g}\Big)\frac{s_{\hat{e},g}^2}{m_g}. \tag{7}$$

Proof sketch. By (i), the Horvitz–Thompson total of any bounded variable has design variance at most a constant times $\sum_g N_g^2/m_g = O(N^2/m)$; hence $\hat{H}_\lambda/N_H \to 1$ in probability and $\hat{t}_v - t_v = O_p(Nm^{-1/2})$ for bounded v. For fixed $(\eta,\lambda) = (\eta^*,\lambda^*)$, (6) is exact and the claim follows from the central limit theorem for $\hat{t}_{e^*}$ and Slutsky's lemma; replacing μ_H by $\hat{\mu}$ in the residual changes $\hat{V}$ by a factor $1 + o_p(1)$, again by (i) and the non-degeneracy in (ii). For fitted quantities, $\hat{T}_\lambda - \mu_H\hat{H}_\lambda = \hat{t}_u - \hat{\lambda}\{\hat{t}_{v(\tilde{\eta})} - t_{v(\bar{\eta})}\}$. The first term does not involve the predictor. By (iv) the bracket equals $\hat{t}_{v(\eta^*)} - t_{v(\eta^*)} + o_p(Nm^{-1/2})$, and since that difference is $O_p(Nm^{-1/2})$, replacing $\hat{\lambda}$ by λ^* costs $o_p(Nm^{-1/2})$ by (iii). Consistency of the variance estimator at the fitted residual follows from (i) and (iii). □

Conditions (ii) and (iv) are assumptions, not consequences of (iii). Condition (ii) holds if e^* satisfies a Lindeberg-type condition in the strata that are not fully enumerated (Bickel and Freedman, 1984, for stratified sampling without replacement); it fails in the degenerate case of a perfect predictor ($e^* \equiv 0$), and it is strained when the residual is non-zero for only a

handful of records, the situation Remark 1 addresses. The allocation condition in (i) holds under proportional allocation, where $m\,N^{-2}\sum_g N_g^2/m_g = 1$; without it, a stratum with a fixed number of audited records and a non-vanishing share of the response set keeps $\hat{H}_\lambda/N_H$ random in the limit, and the limit distribution is not normal. Condition (iv) is the cross-fitting requirement of prediction-powered inference with a fitted predictor (Zrnic and Candès, 2024; Angelopoulos et al., 2023, for the tuning constant), written for the present design; we assume it rather than derive it, and regard it as plausible for a low-dimensional parametric model such as the two-covariate logistic model used here. The part about "both sums" matters: a predictor that interpolates its training labels, applied in the first-phase sum to the audited records it was trained on, violates (iv) and biases the estimate although (iii) holds. Our implementation therefore uses, for an audited record, its out-of-fold prediction in both sums, and the average of the fold models for unaudited records. Folds are balanced within design strata and origin classes. We do not claim the theorem for the two-stage or systematic designs that are compared as secondary analyses. The theorem is asymptotic; with 18 to 90 audited records the coverage of the reported intervals is an empirical question, answered by endpoint PE2.

Corollary 1 (Power tuning is asymptotically no worse than the audit-only estimator)**.** *For fixed η with $V_v = \mathrm{Var}_p(\hat{t}_v) > 0$, the design variance of $\hat{t}_u - \lambda\hat{t}_v$ is $V_u - 2\lambda C_{uv} + \lambda^2 V_v$, minimised at $\lambda^* = C_{uv}/V_v$ with value $V_u(1-\varrho_{uv}^2)$; for λ^* clipped to $[0,1]$ the value is still at most V_u, the variance at $\lambda = 0$. Hence, under the conditions of Theorem 2, a power-tuned member of the family has asymptotic variance no larger than the two-phase Hájek estimator, for any predictor.*

Proof. A convex quadratic in λ restricted to $[0,1]$ attains its minimum at the clipped vertex, and $0 \in [0,1]$. □

The statement is asymptotic: in a small audit the tuning constant is itself estimated, and with an uninformative predictor the tuned estimator can be somewhat worse than the Hájek estimator. No ordering holds between RC-PPI++ and flag-PPI, which use different predictors; their comparison is empirical (endpoint PE1). In the implementation $\hat{\lambda} = \hat{C}_{uv}/\hat{V}_v$ is computed from the audit with μ_H replaced by the $\lambda = 1$ estimate, clipped to $[0,1]$, and set to 1 when $\hat{V}_v = 0$.

Remark 1 (Finite-sample interval)**.** The residuals of a good predictor are sparse: $\hat{e}_i$ is close to zero unless the predictor errs on record i. In a stratum whose few audited records are all predicted correctly, the sample variance $s^2_{\hat{e},g}$ is near zero although the stratum may contain errors. We therefore (a) shrink each stratum variance towards the variance anticipated under a working model in which the binary origins of the audited records of the stratum are independent Bernoulli(p_g) variables. Writing $\hat{e}_i = z_i^0 - a_i\delta_i$ with $\delta_i = y_i - \hat{\mu}$ and treating z^0 and δ as fixed,

$$\mathbb{E}_\xi\big[s^2_{\hat{e},g}\big] = p_g(1-p_g)\,\overline{\delta^2}_g + s^2_{z^0 - p_g\delta,\,g}, \tag{8}$$

which follows from $\mathbb{E}\sum_i(\hat{e}_i - \bar{\hat{e}})^2 = (1-\frac{1}{m_g})\sum_i \delta_i^2 p_g(1-p_g) + \sum_i(c_i - \bar{c})^2$ with $c_i = z_i^0 - p_g\delta_i$. We evaluate (8) at the Jeffreys estimate $p_g = (k_g + \frac{1}{2})/(m_g+1)$, k_g being the number of machine records among the m_g audited, and use $\{(m_g-1)s^2_{\hat{e},g} + \kappa\,\mathbb{E}_\xi[s^2_{\hat{e},g}]\}/(m_g - 1 + \kappa)$ with $\kappa = 1$. (b) We use a Student quantile with Satterthwaite degrees of freedom $(\sum_g \nu_g)^2 \big/ \sum_g\{\nu_g^2/(m_g-1)\}$, at least 1, where ν_g is the contribution of stratum g to $\hat{V}$. Strata with a single audited record are merged with a neighbouring stratum before the variance is computed. This rule is a heuristic selected on development data; its justification is its measured coverage.

1.5 Audit design

RC-PPI needs audited records of both classes to fit its predictor. Let K be the number of audited machine records.

Proposition 3 (Audit starvation)**.** *Let origins be binary. Under simple random sampling of m records, K is hypergeometric with parameters (N, N_A, m) and mean $m\varepsilon$. Under a stratified design with m_g records from a stratum with machine share ε_g, $\mathbb{E}[K] = \sum_g m_g\varepsilon_g$, which equals $m\varepsilon$ under proportional allocation.*

For $N = 900$, $m = 45$ and $\varepsilon = 0.05$, $\Pr(K \geq 4) = 0.18$ and $\Pr(K \geq 6) = 0.02$: at low contamination a small audit rarely contains enough machine records, whatever estimator is used afterwards. Only oversampling high-score strata raises $\mathbb{E}[K]$, and the next result shows what that costs.

Proposition 4 (Allocation)**.** *Under stratified simple random sampling with $\sum_g m_g = m$, ignoring integer constraints and $m_g \leq N_g$, the design variance of $\hat{t}_e$ is $\sum_g N_g^2\sigma_g^2/m_g - \sum_g N_g\sigma_g^2$, where σ_g^2 is the variance of e in stratum g (divisor $N_g - 1$). It is minimised by $m_g \propto N_g\sigma_g$, with minimum $(\sum_g N_g\sigma_g)^2/m - \sum_g N_g\sigma_g^2$; proportional allocation exceeds the minimum by $(N/m)\sum_g N_g(\sigma_g - \bar{\sigma})^2$ with $\bar{\sigma} = \sum_g N_g\sigma_g/N$.*

Proof. By the Cauchy–Schwarz inequality $(\sum_g N_g\sigma_g)^2 \leq (\sum_g N_g^2\sigma_g^2/m_g)(\sum_g m_g)$, with equality if and only if $m_g \propto N_g\sigma_g$. Under proportional allocation the first term is $(N/m)\sum_g N_g\sigma_g^2$, and $\sum_g N_g\sigma_g^2 - (\sum_g N_g\sigma_g)^2/N = \sum_g N_g(\sigma_g - \bar{\sigma})^2$ (Cochran, 1977, Section 5.5). □

The optimal allocation follows the spread of the residual, not the score. Under the working model of Remark 1, the residual spread of a stratum is largest where the origin is uncertain (p_g near one half) and outcomes are dispersed; a top score decile in which nearly every record is machine-written and correctly predicted is not such a place. Because σ_g depends on unverified origins, an allocation close to Neyman's must be learned from a pilot audit, and then the following applies.

Lemma 2 (Two-stage audits)**.** *Let a pilot S_1 be drawn by stratified simple random sampling with sizes $n_{1g} < N_g$, let the second-stage sizes $m_{2g} \geq 1$ be any function of S_1 and of the origins verified in it, and let S_2 be a simple random sample of m_{2g} of the $N_g - n_{1g}$ records of stratum g outside S_1. For any z not depending on the samples,*

$$\hat{t}_z^{(2)} = \sum_{i \in S_1} z_i + \sum_g \frac{N_g - n_{1g}}{m_{2g}} \sum_{i \in S_2 \cap g} z_i \tag{9}$$

satisfies $\mathbb{E}[\hat{t}_z^{(2)} \mid S_1] = t_z$. Consequently: (a) $\hat{t}_z^{(2)}$ is unbiased and $\mathrm{Var}(\hat{t}_z^{(2)}) = \mathbb{E}[\mathrm{Var}(\hat{t}_z^{(2)} \mid S_1)]$, so the stratified variance estimator applied to the second stage alone is unbiased for it when $m_{2g} \geq 2$; (b) the single-stage weights $N_g/(n_{1g} + m_{2g})$ applied to $S_1 \cup S_2$ are in general biased when m_{2g} depends on the pilot origins; (c) when the sizes m_{2g} are fixed in advance, the variance of the stratum-g term of $\hat{t}_z^{(2)}$ is $(N_g - n_{1g})(n_{1g} + m_{2g})/(N_g m_{2g})$ times that of a single-stage simple random sample of $n_{1g} + m_{2g}$ records.

Proof. Given S_1, the sizes m_{2g} are fixed and the second sum is the stratified expansion estimator of $\sum_{i \notin S_1} z_i$ on the population $U \setminus S_1$; (a) follows since the conditional mean is the constant t_z. For (b) take one stratum with $N = 3$, $z = a = (1, 0, 0)$, $n_1 = 1$, and $m_2 = 1$ if the pilot record is machine-written and $m_2 = 2$ otherwise: the single-stage estimator has expectation $\frac{1}{3} \cdot \frac{3}{2} \cdot 1 + \frac{2}{3} \cdot 1 \cdot 1 = \frac{7}{6} \neq 1$. For (c), with σ^2 the stratum variance of z (divisor $N - 1$) and R the $N - n_1$ records outside the pilot, the conditional variance is $(N - n_1)(N - n_1 - m_2)\sigma_R^2/m_2$ and $\mathbb{E}[\sigma_R^2] = \sigma^2$ because R is itself a simple random sample; a single-stage sample of $n_1 + m_2$ has variance $N(N - n_1 - m_2)\sigma^2/(n_1 + m_2)$. □

Part (c) is the price of adaptivity: pilot records count only for themselves, and with a pilot of 30% of the budget the factor is about $1/0.7 = 1.4$ before any gain from a better allocation. Whether a label-adaptive design nevertheless beats proportional allocation, whose weights are

known in advance and whose loss relative to Neyman allocation is given in Proposition 4, is an empirical question; at the budgets and with the allocation rules studied here it did not on the development corpora, and the comparison is repeated on the held-out corpora as a secondary analysis.

1.6 Certification, verification error, and regret

Let $b = \bar{y}_{d=0} - \mu_H$ be the deletion bias on the response set and $\hat{b} = \bar{y}_{d=0} - \hat{\mu}$ its estimate from an audit-based $\hat{\mu}$ with standard error $\widehat{\mathrm{se}}$. Since $\bar{y}_{d=0}$ is computed from the whole response set, $\hat{b} - b = -(\hat{\mu} - \mu_H)$.

Proposition 5 (Certifying deletion)**.** *Fix a tolerance $\tau > 0$ and certify deletion when $|\hat{b}| + q_{1-\alpha}\,\widehat{\mathrm{se}} < \tau$, with $q_{1-\alpha}$ at least the standard normal $(1-\alpha)$ quantile. If $(\hat{\mu} - \mu_H)/\widehat{\mathrm{se}}$ is asymptotically standard normal, then $\limsup \Pr(\textit{certify}) \le \alpha$ whenever $|b| \ge \tau$.*

Proof. If $b \ge \tau$, certification implies $\hat{b} + q_{1-\alpha}\widehat{\mathrm{se}} < \tau \le b$, that is $(\hat{\mu} - \mu_H)/\widehat{\mathrm{se}} > q_{1-\alpha}$, an event of limiting probability at most α. The case $b \le -\tau$ is symmetric. □

This is the two one-sided tests procedure for equivalence. We use $\tau = 0.2$ standard deviations of the outcome in the response set, a substantive choice, and a Student quantile with the degrees of freedom of Remark 1. The audit-based estimate is the complete rule described after Theorem 1 (RC-PPI++, or flag-PPI when RC-PPI++ is infeasible); the normal limit is stated above, under the assumptions of Theorem 2, for each estimator separately but not for the rule that switches between them on an origin-dependent event, so the guarantee is asymptotic and conditional on that limit, and the false-certification rate in small audits is measured (endpoint PE3). The rule of the first submission compared a bootstrap interval of $\hat{b}$ with zero and had no such guarantee.

Proposition 6 (Verification error)**.** *Let origins be binary and suppose the audit returns $\tilde{a}_i$ instead of a_i, with independent errors $\Pr(\tilde{a}_i = 1 \mid a_i = 0) = \varphi_0 < 1$ and $\Pr(\tilde{a}_i = 0 \mid a_i = 1) = \varphi_1$ that do not depend on y_i or on the design. For fixed (η, λ), $\mathbb{E}[\hat{T}_\lambda]/\mathbb{E}[\hat{H}_\lambda] = \tilde{\mu}$ with*

$$\tilde{\mu} - \mu_H = \frac{\varphi_1\,\varepsilon\,(\mu_A - \mu_H)}{(1-\varphi_0)(1-\varepsilon) + \varphi_1\varepsilon}. \tag{10}$$

Proof. $\mathbb{E}[1 - \tilde{a}_i] = (1-\varphi_0)h_i + \varphi_1 a_i$. Repeating part (1) of Theorem 1 with this in place of h_i gives $\mathbb{E}[\hat{T}_\lambda] = (1-\varphi_0)N_H\mu_H + \varphi_1 N_A\mu_A$ and $\mathbb{E}[\hat{H}_\lambda] = (1-\varphi_0)N_H + \varphi_1 N_A$. □

The quantity $\tilde{\mu}$ is the ratio of the expectations, to which the estimator is close in large audits. Without machine records passed as human ($\varphi_1 = 0$) it equals μ_H: humans wrongly rejected (φ_0) alone cost precision, not validity, although they amplify the shift when $\varphi_1 > 0$. Its distance from μ_H is at most $\varphi_1 \frac{\varepsilon}{1-\varepsilon}|\mu_A - \mu_H|/(1-\varphi_0)$, which gives a sensitivity bound for stated bounds on φ_0, φ_1, ε and the range of the outcome; these cannot be estimated from the erroneous audit itself.

Definition 2 (Regret)**.** A *cell* is a corpus, detector, contamination rate, audit budget and outcome. For an estimation rule r let $\mathrm{sRMSE}_r(c)$ be its root mean squared error in cell c over replications, divided by the standard deviation of the human outcome. The regret of r in c is $\mathrm{sRMSE}_r(c) - \min_{r' \in \mathcal{R}} \mathrm{sRMSE}_{r'}(c)$, where $\mathcal{R}$ is the set of single estimators (naive, deletion, two-phase Hájek, flag-PPI, RC-PPI++). The comparator is the best single estimator in hindsight, which no analyst can select without the truth; regret is non-negative for single estimators and may be negative for a rule that switches between them.

Remark 2 (Selection policy)**.** The policy reports the candidate (naive, deletion, or the audit-based estimate just described) with the smallest estimated mean squared error. For candidate $j \in \{\text{naive}, \text{deletion}\}$ the error is its bias b_j on the response set, estimated by $\hat{b}_j = \hat{\mu}_j - \hat{\mu}$; since $\mathbb{E}[\hat{b}_j^2] \approx b_j^2 + \mathrm{Var}(\hat{\mu})$, the plug-in $\hat{b}_j^2$ overstates the squared bias and the policy uses $\max(\hat{b}_j^2 - \widehat{\mathrm{se}}^{\,2}, 0)$,

compared with $\widehat{\mathrm{se}}^{\,2}$ for the audit-based estimate. Equivalently, naive or deletion is preferred to the audit-based estimate only if $|\hat{b}_j| < \sqrt{2}\,\widehat{\mathrm{se}}$. If both have estimated squared bias zero, the naive mean is reported. The interval reported with the policy is that of the audit-based estimate. Note the two scales: the certification tolerance is in standard deviations of the outcome over all records of the response set, which the analyst observes, whereas regret is standardised by the human standard deviation, which only the evaluation knows.

Numerical checks

Statement	Test (repository `tests/theory/`)
Lemma 1, Proposition 2	`test_identities.py`: exact to 10^{-10} on response sets
Proposition 1	`test_exact_statements.py`: the solution equals μ_H on a response set built to satisfy M1 exactly
Theorem 1	`test_exact_statements.py`: unbiasedness by enumeration (simple and stratified sampling, graded origins, λ outside $[0, 1]$); representation (6) exact for the package estimators
Theorem 2	`test_exact_statements.py`: coverage of the linearised normal interval (7) in a large audit (simulation); `test_design_based.py`: reported interval
Corollary 1	`test_exact_statements.py`: estimated variance at the package's $\hat{\lambda}$ never above that at $\lambda = 0$, stratified audits
Remark 1	`test_exact_statements.py`: (8) by enumeration of all label vectors
Proposition 3	`test_exact_statements.py`: both probabilities
Proposition 4	`test_exact_statements.py`: minimum against random allocations; excess of proportional allocation
Lemma 2	`test_exact_statements.py`: (a)–(c) by enumeration; `tests/unit/test_audit_designs.py`: unbiased totals for the package's two-stage designs (simulation)
Proposition 5	`test_exact_statements.py`: certification rate at the boundary $\lvert b\rvert = \tau$ not above α (simulation, one-sided 1% test)
Proposition 6	`test_exact_statements.py`: (10) exact; `test_design_based.py`: simulation
Remark 2	`test_exact_statements.py`: the package policy follows the $\sqrt{2}$ rule